

An open-source stack for in-situ heliophysics

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In-situ data is **spread over many archives**, each with its own formats, access method and metadata quality. Most tools help you make one plot, then leave you on your own. **SciQLop**, **Speasy**, **pycdfpp** and **AstraLint** cover the whole path, from **raw mission files** to a **result you can reproduce**.

Four tools, each useful on its own

Speasy — one API for every archive

- **70+ missions, 77 000+ products**
- AMDA, CDAWeb, CSA, SSCWeb, CDPP 3DView, plus your own file archives
- A **local cache**, and a **shared public one** hosted at LPP
- Returns a `SpeasyVariable`: time, values and metadata, usable as a **NumPy array**

v1.8.1 · MIT · pip install speasy



pycdfpp — fast CDF, everywhere

- **Reads and writes** NASA CDF without the NASA library
- **Zero-copy** NumPy access, reads at up to **~5 GB/s**
- Wheels for Linux, macOS, Windows and **WebAssembly**
- **Try it in your browser**: sciqlop.github.io/CDfpp

v0.11.1 · MIT · pip install pycdfpp



SciQLop — the in-situ space physics IDE

- Browse **any product from any mission**, at full resolution
- **Embedded JupyterLab**: every plot can be scripted
- **Virtual products**: your Python function, recomputed as you pan and zoom
- **Event catalogues, plugins** and a built-in **app store**

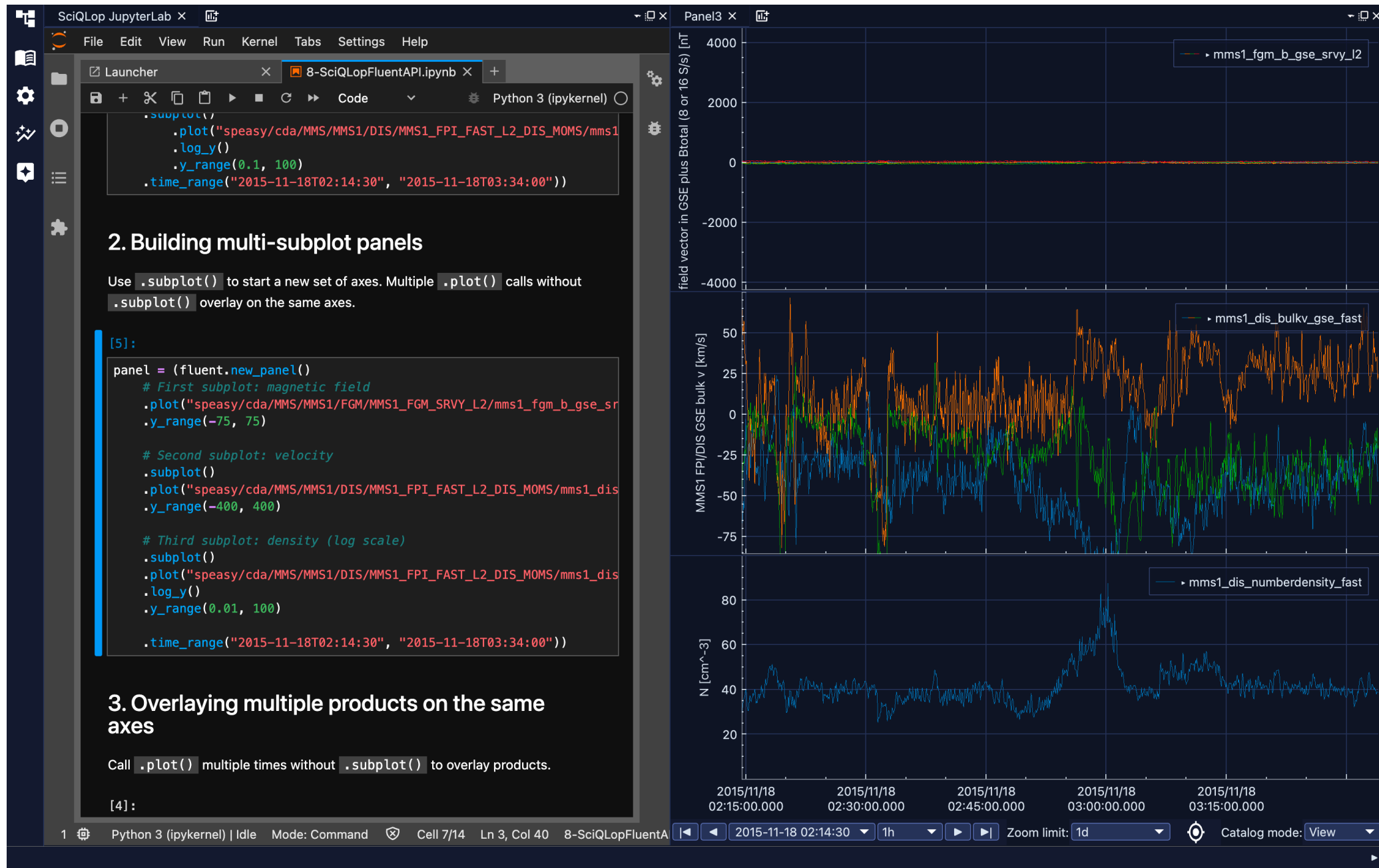
v0.13.0 · GPLv3 · installers for Linux, macOS and Windows



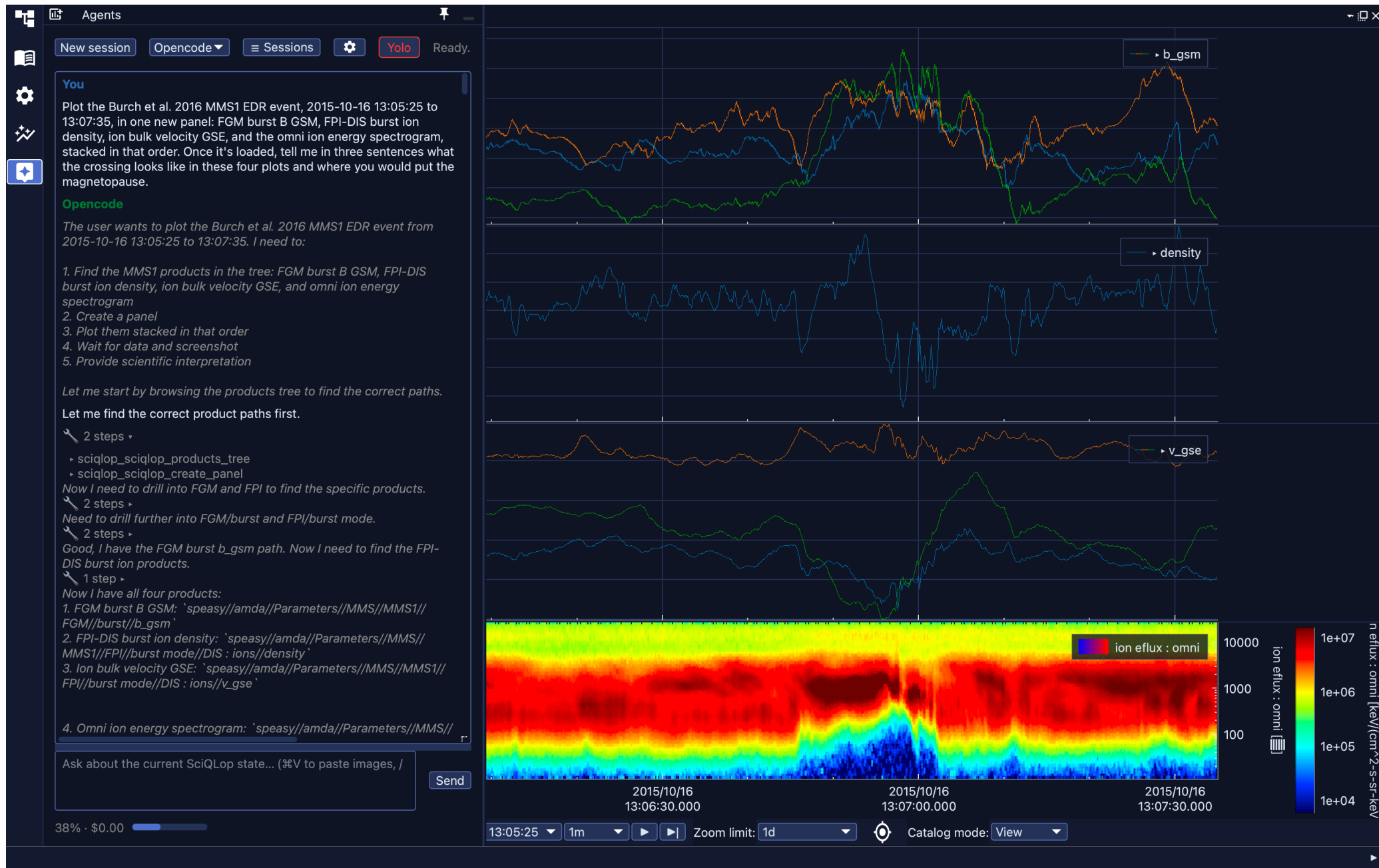
AstraLint — quality control at the source

- Checks CDF metadata against the **ISTP** and **PDS4** conventions
- `astralint fix` applies the repairs that are **safe to automate**
- Rules are written in YAML or Python: **add your mission's own**
- Command line, CI, or **online**: sciqlop.github.io/AstraLint

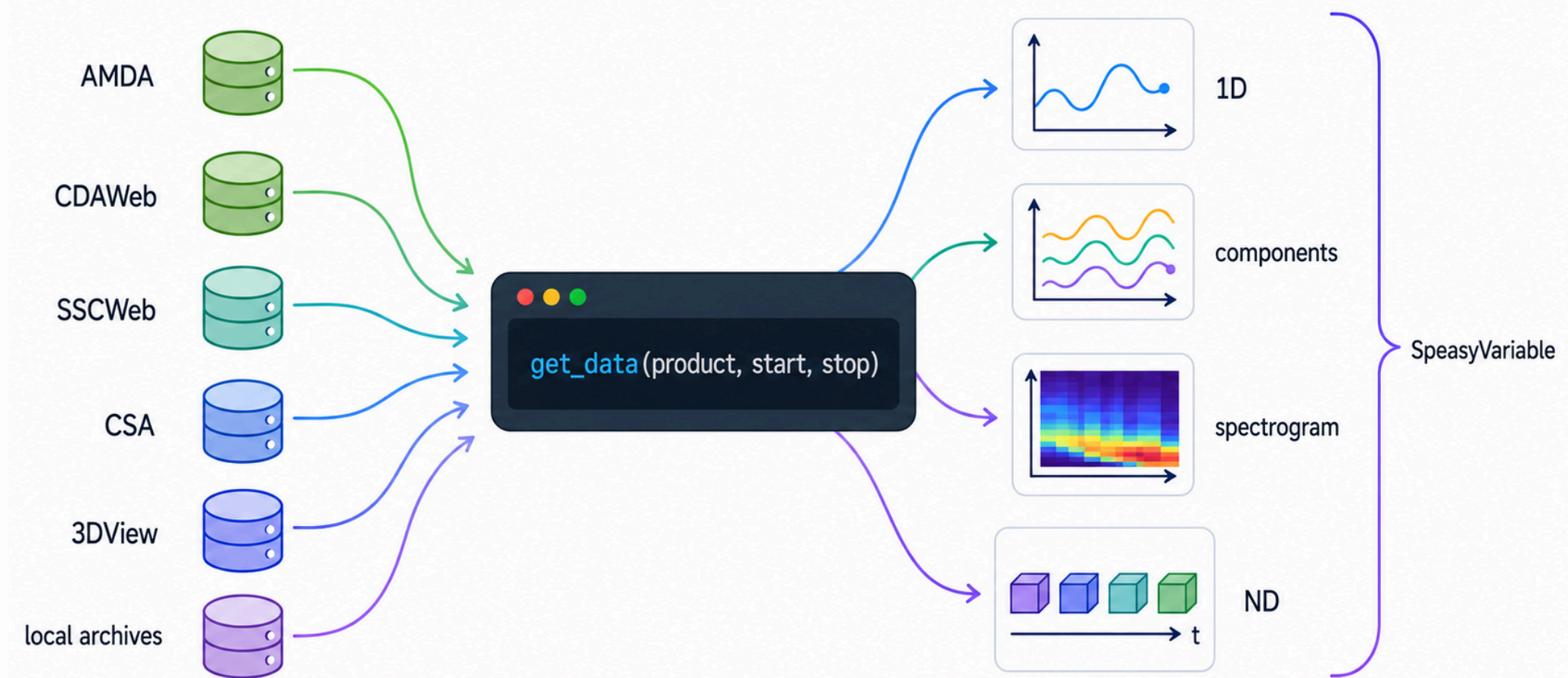
v0.9.1 · MIT · pip install astralint



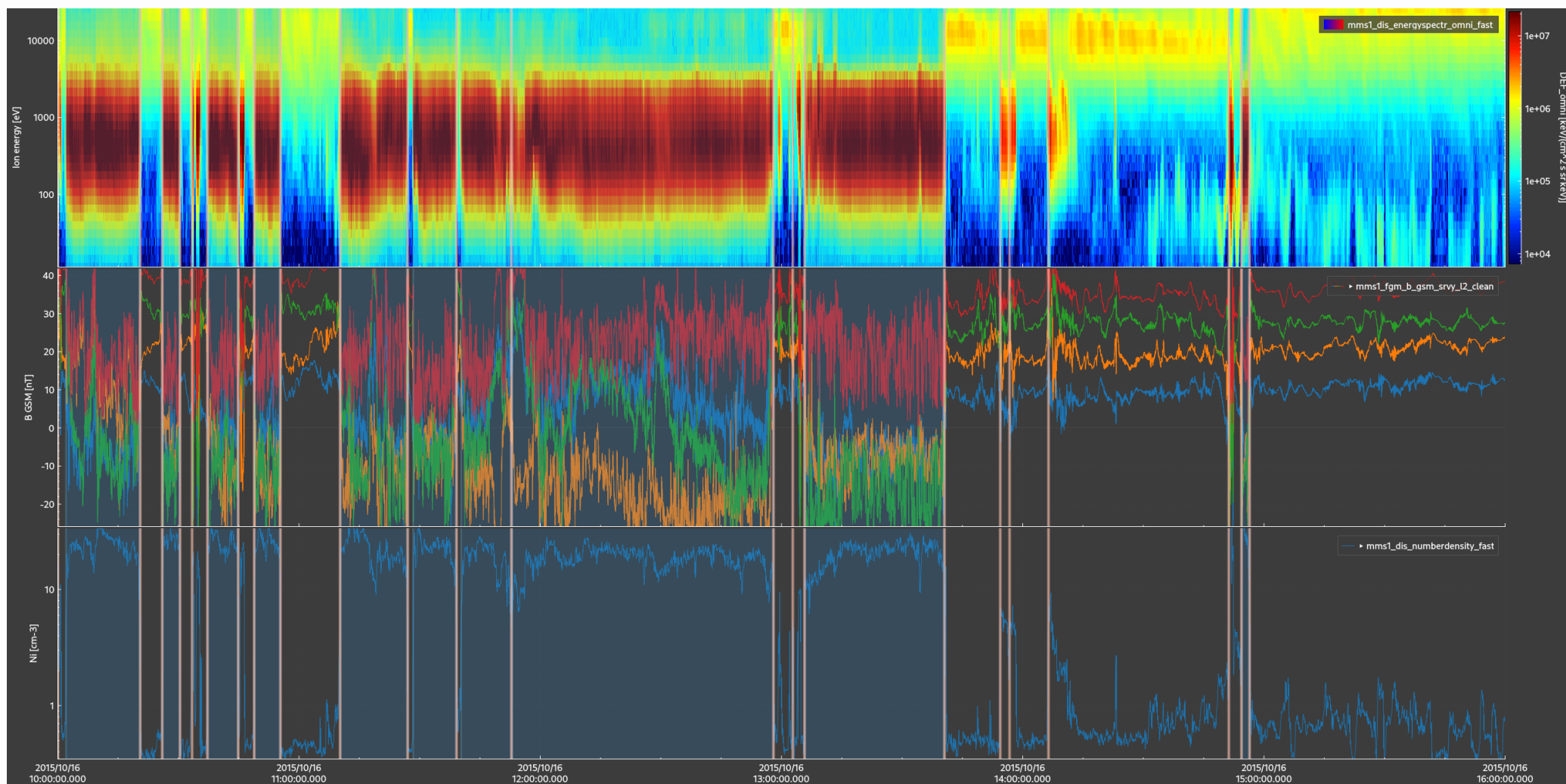
SciQLop: a notebook in the embedded JupyterLab builds the plot panel beside it



SciQLop: the AI assistant finds the products and builds the panel from one sentence



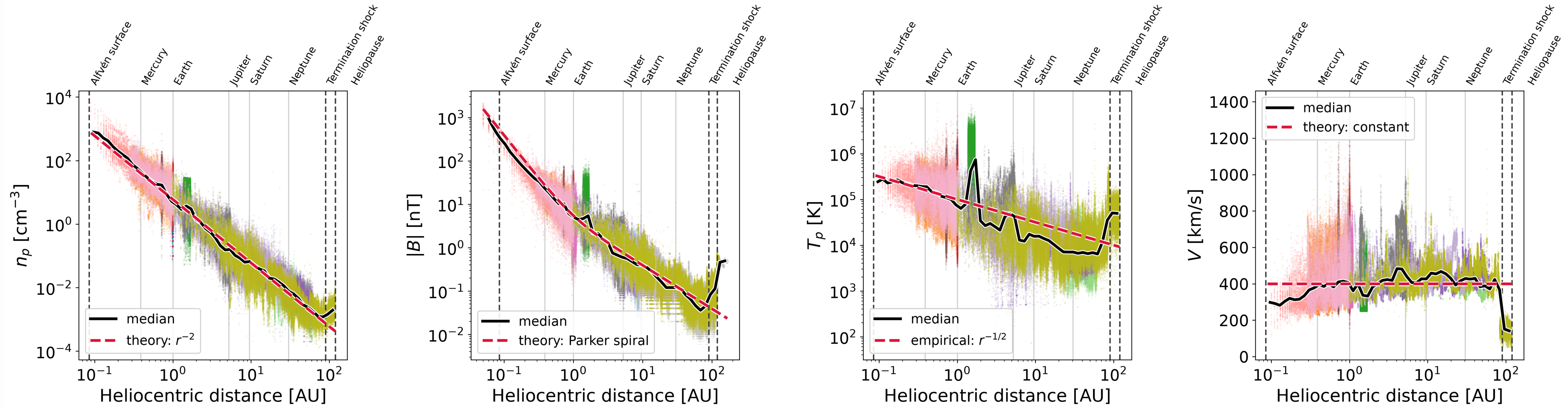
Speasy: every archive behind one call, one return type



SciQLop: event catalogues drawn over the data

Worked example: a whole-heliosphere solar-wind survey

17 missions in one table: 0.05 to 167 AU, 1972 to 2026, 1.76 M hourly samples.



- | | | | | | | | | |
|-------------|------------|----------------|---------------|--------------|-----------------|-----------------|-------------|-------------|
| ● Aditya-L1 | ● Helios 1 | ● MAVEN | ● OMNI (1 AU) | ● Pioneer 10 | ● Pioneer Venus | ● STEREO-B | ● Ulysses | ● Voyager 2 |
| ● Cassini | ● Helios 2 | ● New Horizons | ● PSP | ● Pioneer 11 | ● STEREO-A | ● Solar Orbiter | ● Voyager 1 | |

Every hourly sample, coloured by mission, with the median (black) and the expected law (dashed). Vertical lines: planets and heliospheric frontiers.

17 missions

Parker Solar Probe · Helios 1/2 · Solar Orbiter · STEREO-A/B · Pioneer Venus Orbiter · OMNI · Aditya-L1 · MAVEN · Ulysses · Pioneer 10/11 · Voyager 1/2 · New Horizons · Cassini

What the stack did

- Speasy fetched **16 of the 17 missions**, from CDAWeb and AMDA
- **One table** maps each mission's names onto common columns
- Thanks to the **cache**, a re-run downloads nothing twice
- Built in **SciQLop's notebooks**, next to the plots

What came out

546 times, two spacecraft saw the **same parcel** of solar wind. Between them $|B|$ falls as $r^{-1.43}$, where the Parker spiral gives about $r^{-1.5}$, and the speed does not change, as expected. **Getting the data was the easy part.**